COLLECTIVE - Literature-Grounded Feasibility Review

Can We Build Artificial Intelligence from Programmable Living Units?
A critical comparison with peer-reviewed research
Version 0.2 | 12 November 2026

RESEARCH STATUS
Exploratory, speculative, and unvalidated. The cited studies demonstrate components or related phenomena, not the proposed integrated COLLECTIVE system. The purpose is to identify testable interfaces and decisive gaps, not to claim novelty or feasibility prematurely.

EXECUTIVE SUMMARY

COLLECTIVE proposes heterogeneous programmable biological or biohybrid units with addressable DNA-based information storage, local electronic processing and RAM, calibrated sensors, standardized communications, metabolism, and a speculative processor-directed molecular manufacturing interface. The units would learn cooperation and task allocation, with emergent organization and physical innovation treated as hypotheses to test.

Literature supports selected DNA-file retrieval [R1-R4], electronic-biological interfacing [R5], stimulus-responsive synthetic-cell protein production [R6-R8,R12], closed-loop adaptation in cultured neurons [R9], and self-organized behavior in living cell assemblies [R10-R11]. None demonstrates the full architecture or general organic AI.

The most decisive gaps are random-access DNA readout compatible with practical processor execution, general-purpose electronic control of molecular synthesis, stable metabolism/material flow, credible attribution of cooperative learning, and tests that separate emergent innovation from programmed optimization.

1. DNA AS AN ADDRESSABLE PROGRAM ARCHIVE

ESTABLISHED: Organick et al. recovered individually selected files from DNA storage using molecular addressing [R1]. Reviews describe progress in selective retrieval and continuing bottlenecks including latency, amplification bias, cross-talk, and access scalability [R2-R4].

COLLECTIVE EXTENSION: indexed DNA records contain executable instruction sequences and reference data. An electronic processor locates a record, reads and decodes it, places instructions into RAM, and executes them (for example, an FFT routine).

GAP: addressability in DNA storage is not equivalent to low-latency, high-endurance electronic memory. Molecular readout can be slow and may involve destructive or batch processing. Boot logic, ISA, decoding, read bandwidth, error correction, and physical coupling remain to be specified.

FIRST TEST: an explicit latency/energy/error budget for retrieval of a small encoded routine, benchmarked against a conventional non-DNA archive. Failure to meet a defined threshold is informative.

2. BIOELECTRONIC AND SENSOR INTERFACES

ESTABLISHED: Biohybrid platforms combine engineered cell responses and electronic readout or control; a recent review maps communication mechanisms, applications, and integration limits [R5].

COLLECTIVE EXTENSION: units have standardized ports for optical, chemical, electrochemical, mechanical, and pH measurements; signal processing is local or distributed over a predefined serial bus.

GAP: living sensors drift and respond to context. Electronic and biochemical timescales differ; calibration, noise, selectivity, communication bandwidth, power, and wet-environment compatibility matter. A serial bus is an engineering assumption, not an existing cell organelle.

FIRST TEST: compare a calibrated sensor-to-processor interface with independent instruments under bounded noise and drift; model asynchronous communication overhead before biology is involved.

3. MOLECULAR MANUFACTURING AND SYNTHETIC CELLS

ESTABLISHED: Temperature-responsive synthetic cells have been shown to regulate protein synthesis and cargo release using encapsulated expression machinery [R6]. Synthetic-cell and cell-free expression reviews describe advances, and emphasize the continuing gap between component construction and fully functional synthetic life [R7,R8,R12].

COLLECTIVE EXTENSION: a processor specifies molecular products and receives feedback on synthesis, folding, assembly, and downstream chemical effects; the long-term concept goes beyond conventional ribosomal production.

GAP: an electronic sequence specification does not physically synthesize a molecule. Precursor supply, energy, catalysts, translation or alternative chemistry, folding, localization, quality control, and waste management are separate hard problems. General ribosome replacement is not demonstrated by stimulus-controlled expression.

FIRST TEST: develop a requirements matrix for each manufacturing stage and identify which stages already have independent experimental demonstrations. No general synthesis claim should be made without end-to-end evidence.

4. BIOLOGICAL ADAPTATION AND COLLECTIVE ORGANIZATION

ESTABLISHED: DishBrain demonstrated feedback-dependent adaptive behavior in cultured neural networks in a constrained game environment [R9]. Xenobots demonstrate self-organization, coordinated movement, and group behavior in living cellular constructs [R10]. Bioelectric signaling literature describes how cellular networks coordinate large-scale anatomical outcomes [R11].

COLLECTIVE EXTENSION: non-identical units share tasks, adapt communication weights, and potentially develop specializations and network hierarchy while solving diverse problems.

GAP: neuronal adaptation, morphological self-organization, and multi-agent optimization are different phenomena. These precedents do not show that arbitrary biohybrid units will acquire general intelligence or invent novel hardware. Claims about sentience in paper titles should not be imported as established conclusions.

FIRST TEST: compare heterogeneous learning agents with matched centralized, fixed-rule, and randomized baselines on unseen tasks; preregister operational metrics of generalization, resource efficiency, and cooperative gain.

5. REWARD AND CREDIT ASSIGNMENT

COLLECTIVE proposes both individual scores and collective outcomes. Reward may reflect accuracy, verified task completion, speed, computation contributed, and resource cost. A unit that consumes more computation may appear more deserving under a raw-work metric, but that can encourage waste. An efficient unit may finish the same task faster with fewer operations.

This is a mechanism-design hypothesis, not a validated biological reward system. Compare multiple reward policies, including work-weighted, marginal-contribution, efficiency-adjusted, and team-only variants. Record redundant computation, strategic delay, message flooding, free-riding, and changes in network centrality.

The question is not merely whether hierarchy emerges, but whether any hierarchy improves held-out task performance or exploits the scoring rule.

6. PHYSICAL INNOVATION AND NOVELTY

COLLECTIVE deliberately does not supply explicit goals such as grow an organ, move, add hardware, or replicate. Instead, it asks whether problem-solving units could discover useful intermediate physical capabilities, provided their interfaces permit some design, fabrication, and evaluation.

A novel-looking outcome may still be a recombination of stored designs, a consequence of a generic optimizer, or an artifact of the evaluator. Therefore maintain a complete inventory of preloaded programs and permitted actions. Test unseen constraints, blinded artifact evaluation, and performance relative to ablated agents.

This remains a long-range experimental hypothesis. No cited study demonstrates this outcome for the proposed architecture.

7. COMPARATIVE FEASIBILITY MATRIX

Subsystem | Literature-supported baseline | COLLECTIVE target | Current assessment
DNA retrieval | selective molecular file access [R1-R4] | executable random-access archive | major interface gap
Bioelectronics | sensing and control links [R5] | standardized multi-sensor unit | partial precedent
Synthetic cells | regulated expression/cargo release [R6-R8] | general processor-directed synthesis | very large gap
Metabolism | components in living and cell-free systems [R7,R12] | integrated sustained resource economy | major gap
Adaptive behavior | constrained neural adaptation [R9] | transferable multi-unit learning | unproven integration
Self-organization | cellular assembly and bioelectric patterning [R10,R11] | open-ended physical innovation | speculative

These qualitative judgments are provisional and are not technology-readiness-level ratings. The literature search is selective, not a systematic review.

8. PRIORITIZED RESEARCH PROGRAM

PHASE A: Formalize terms and baselines. Specify intelligence, novelty, adaptation, collective advantage, and failure criteria. Publish a feasibility matrix with verified

citations.

PHASE B: Digital experiments. Simulate heterogeneous units with limited compute, energy, memory, and bus bandwidth. Compare rewards and topology adaptation. Test unseen tasks and adverse incentives.

PHASE C: Quantitative interface review. Estimate DNA access latency, throughput, read errors, energy, and cost. Evaluate sensor bandwidth and chemical feedback. Map the material and energy balance of molecular production.

PHASE D: Independent, institutionally supervised component validation if justified by prior evidence and safety review. Avoid claiming that success of one interface implies an integrated living AI system.

PHASE E: Decide whether integration is justified, and publish negative results. Investigate emergence only after a reproducible platform exists.

9. LIMITATIONS AND INVITATION

This report is not a systematic literature review; it draws on a selected set of verified peer-reviewed papers and reviews. It does not establish that the overall COLLECTIVE combination is unique, patentable, safe, or achievable. A comprehensive prior-art search, expert assessment, and formal evidence grading are needed before strong novelty claims.

Potential collaborators: synthetic biology, bioelectronics, DNA data storage, molecular manufacturing, microbiology, artificial life, distributed learning, computer architecture, control theory, materials science, and research ethics.

Contributors are invited to identify an established capability, a missing interface, a quantitative bottleneck, a falsifying experiment, or a simpler alternative architecture.

Repository: https://github.com/altugkuran/COLLECTIVE-organic-AI

CENTRAL QUESTION: Can learned cooperation among programmable living units produce organically embodied artificial intelligence? The answer remains unknown.

REFERENCES

[R1] Organick et al. (2018). Random access in large-scale DNA data storage. Nature Biotechnology. DOI: 10.1038/nbt.4079
https://pubmed.ncbi.nlm.nih.gov/29457795/

[R2] Yu et al. (2026). Random access elevates DNA from a storage medium to a storage system. Trends in Biotechnology. DOI: 10.1016/j.tibtech.2026.06.020
https://www.sciencedirect.com/science/article/pii/S0167779926002799

[R3] Zhou et al. (2024). Advances and Challenges in Random Access Techniques for In Vitro DNA Data Storage. ACS Applied Materials & Interfaces. DOI: 10.1021/acsami.4c07235
https://pubmed.ncbi.nlm.nih.gov/39110103/

[R4] DNA as a universal chemical substrate (2024). DNA as a universal chemical substrate for computing and data storage. Nature Reviews Chemistry. DOI: 10.1038/s41570-024-00576-4
https://www.nature.com/articles/s41570-024-00576-4

[R5] Integrating bioelectronics (2025). Integrating bioelectronics with cell-based synthetic biology. Nature Reviews Bioengineering. DOI: 10.1038/s44222-024-00262-6

https://www.nature.com/articles/s44222-024-00262-6

[R6] Monck, Elani & Ceroni (2024). Genetically programmed synthetic cells for thermo-responsive protein synthesis and cargo release. Nature Chemical Biology. DOI: 10.1038/s41589-024-01673-7
https://www.nature.com/articles/s41589-024-01673-7

[R7] Rothschild (2024). Building Synthetic Cells-From the Technology Infrastructure to Cellular Entities. ACS Synthetic Biology. DOI: 10.1021/acssynbio.3c00724
https://pmc.ncbi.nlm.nih.gov/articles/PMC11037263/

[R8] Samanta et al. (2024). DNA-empowered synthetic cells as minimalistic life forms. Nature Reviews Chemistry. DOI: 10.1038/s41570-024-00606-1
https://www.nature.com/articles/s41570-024-00606-1

[R9] Kagan et al. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. Neuron. DOI: 10.1016/j.neuron.2022.09.001
https://doi.org/10.1016/j.neuron.2022.09.001

[R10] Blackiston et al. (2021). A cellular platform for the development of synthetic living machines. Science Robotics. DOI: 10.1126/scirobotics.abf1571
https://pubmed.ncbi.nlm.nih.gov/34043553/

[R11] Levin (2021). Bioelectric signaling: Reprogrammable circuits underlying embryogenesis, regeneration, and cancer. Cell. DOI: 10.1016/j.cell.2021.02.034
https://pubmed.ncbi.nlm.nih.gov/33826908/

[R12] Cell-Free Gene Expression (2025). Cell-Free Gene Expression: Methods and Applications. Chemical Reviews. DOI: 10.1021/acs.chemrev.4c00116
https://pubs.acs.org/doi/10.1021/acs.chemrev.4c00116